

SPACEX FALCON 9

FIRST STAGE LANDING PREDICTION

PREDICTING WHETHER SPACEX FALCON 9 FIRST STAGE WILL LAND SUCCESSFULLY TO ESTIMATE LAUNCH COSTS

EXECUTIVE SUMMARY

BEST MODEL

LOGISTIC REGRESSION

Best accuracy on test data

DATA SOURCES

SpaceX + Wikipedia

API calls + web scraping

LAUNCH SITES

4 sites analysed

Folium interactive maps

METHODS & KEY RESULTS

- Collected launch data via SpaceX REST API and Wikipedia web scraping (BeautifulSoup)
- Cleaned and wrangled data, mapped landing outcomes to binary labels (1 = success, 0 = failure)
- Performed EDA with Matplotlib/Seaborn charts and SQL queries on SQLite database
- Built interactive Folium map (launch markers + proximity analysis) and Plotly Dash dashboard
- Trained & tuned 4 ML classifiers (Logistic Regression, SVM, Decision Tree, KNN) via GridSearchCV

Github: <https://github.com/IT24103131/spacex-falcon9-landing-prediction.git>



INTRODUCTION

Project Background

SpaceX Falcon 9 launches cost \$62M vs. competitors' \$165M+

**The key to this saving is reusing the first stage booster.
If the first stage lands safely, SpaceX can reuse it,
dramatically cutting launch costs.**

Problem Statement

Can we predict whether the Falcon 9 first stage will land successfully; given launch parameters, so that launch costs can be accurately estimated and competitor bids can be made?

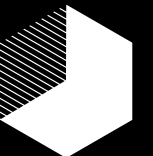
Project Objectives

Collect data via SpaceX API & web scraping

Wrangle & clean raw data into ML-ready format

Analyse with SQL & visualizations

Build & evaluate predictive ML models



DATA COLLECTION – SPACEX API

1 GET / launches

api.spacexdata.com/v4

2 Extract IDs

rocket, launchpad, payload

3 Lookup Endpoints

booster, core, crew

4 Build DataFrame

pandas + cleanup

Key Fields Collected

- FlightNumber, Date, BoosterVersion
- PayloadMass, Orbit, LaunchSite
- Outcome, Flights, GridFins
- Reused, Legs, LandingPad
- Block, ReusedCount, Serial

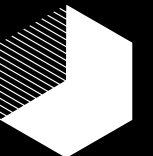
API Call Pattern

```
import requests

url = 'https://api.spacexdata.com/v4/launches'
response = requests.get(url)
data = response.json()

# Extract nested fields
for launch in data:
    getBoosterVersion(data)
    getLaunchSite(data)
```

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DATA COLLECTION – WEB SCRAPING

Source: Wikipedia - List of Falcon 9 and Falcon Heavy Launches

① HTTP Request

`requests.get()` to fetch the Wikipedia HTML page containing launch records table

② Parse HTML

BeautifulSoup parses the page; identifies all wikipables with launch data

③ Extract Rows

Iterate table rows, apply helper functions: `date_time()`, `booster_version()`, `landing_status()`

④ Build DataFrame

Compile extracted fields into structured pandas DataFrame and export to CSV

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DATA WRAGLING & METHODOLOGY

Missing Values

Identified & imputed null PayloadMass values using mean; dropped rows with missing critical fields

Outcome Encoding

Mapped 8 landing outcome types (True/False Ocean, RTLS, ASDS) to binary class: 1 = success, 0 = failure

Feature Selection

Selected relevant features: FlightNumber, PayloadMass, Orbit, LaunchSite, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount

One-Hot Encoding

Applied `pd.get_dummies()` on categorical columns (Orbit, LaunchSite, LandingPad) to create numeric ML features

Landing Outcome → Training Label Mapping

Outcome	True Ocean	False Ocean	True RTLS	False RTLS	True ASDS	False ASDS
Label	1 ✓	0 ✗	1 ✓	0 ✗	1 ✓	0 ✗

EDA – DATA VISUALIZATION

Flight Number vs. Payload Mass

Scatter plot revealed heavier payloads tend to have more controlled outcomes; later flights show higher success rates

Success Rate by Orbit Type

Bar chart shows ES-L1, GEO, HEO, SSO orbits achieve 100% success. LEO and ISS show moderate rates

Launch Success Yearly Trend

Line chart confirms success rate improved significantly after 2013 as SpaceX refined booster recovery technology

Launch Site vs. Flight Number

CCAFS LC-40 had the most early launches; KSC LC-39A emerged as the highest-success site in later years

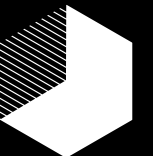
Payload Mass vs. Orbit

Box plots show GTO orbit has widest payload range; polar orbits cluster around lower mass payloads

Success/Failure by Launch Site

Catplot breakdown per site confirms KSC LC-39A has the highest proportion of successful landings

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EDA – SQL ANALYSIS

Launch Sites

```
SELECT DISTINCT launch_site FROM SPACEXTBL
```

→ 4 unique launch sites identified

Payload Range

```
SELECT MIN(payload_mass_kg) AS min, MAX(payload_mass_kg) AS max FROM SPACEXTBL
```

→ 0 kg – 15,600 kg range

Success Rates

```
SELECT launch_site, AVG(mission_outcome) FROM SPACEXTBL GROUP BY launch_site
```

→ KSC LC-39A top performer

Booster Rankings

```
SELECT booster_version, COUNT(*) FROM SPACEXTBL GROUP BY booster_version ORDER BY COUNT(*) DESC
```

→ F9 B5 most frequent booster

Time Analysis

```
SELECT YEAR(date), COUNT(*) FROM SPACEXTBL GROUP BY YEAR(date)
```

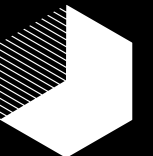
→ Launch frequency grew year-on-year

Landing Pad Success

```
SELECT landing_pad, SUM(class) FROM SPACEXTBL GROUP BY landing_pad
```

→ OCISLY pad had highest successes

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INTERACTIVE VISUAL ANALYTICS

Folium - Launch Site Maps

- Marked all 4 launch sites (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E) on interactive map
- Colour-coded markers: ● green = success, ● red = failure for each site's history
- Calculated distances to nearest city, highway, railway, and coastline using MousePosition plugin
- Proximity analysis revealed all sites are near coastlines (controlled crash zone) and transport links

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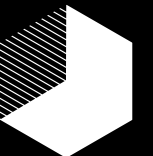
Plotly Dash - Dashboard

- Dropdown filter: select 'All Sites' or a specific launch site to update all charts
- Pie chart: shows total successful launches by site (all sites) or success vs failure breakdown (per site)
- RangeSlider: filter by payload mass 0–10,000 kg to narrow scatter chart data
- Scatter chart: Payload Mass (kg) vs. Launch Outcome coloured by Booster Version Category

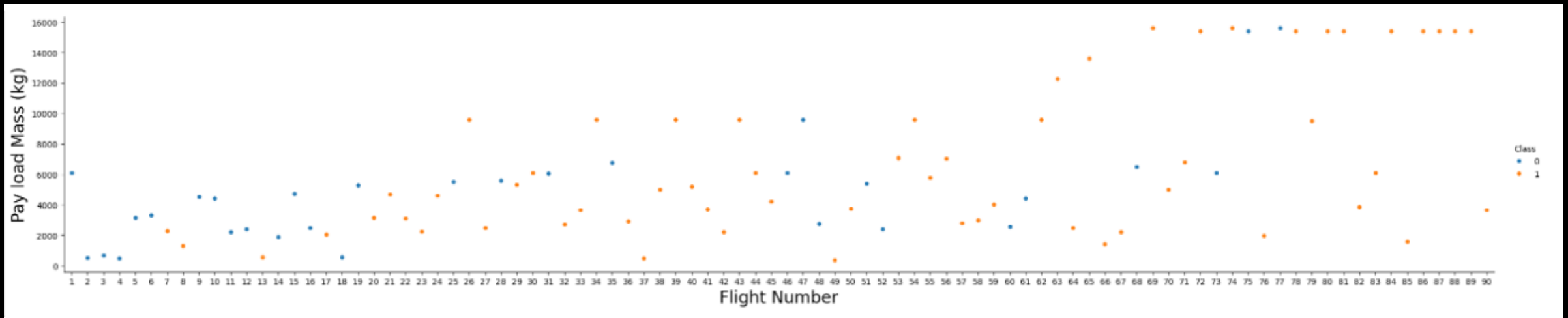
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EDA – DATA VISUALIZATION

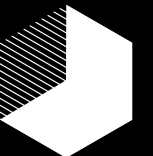
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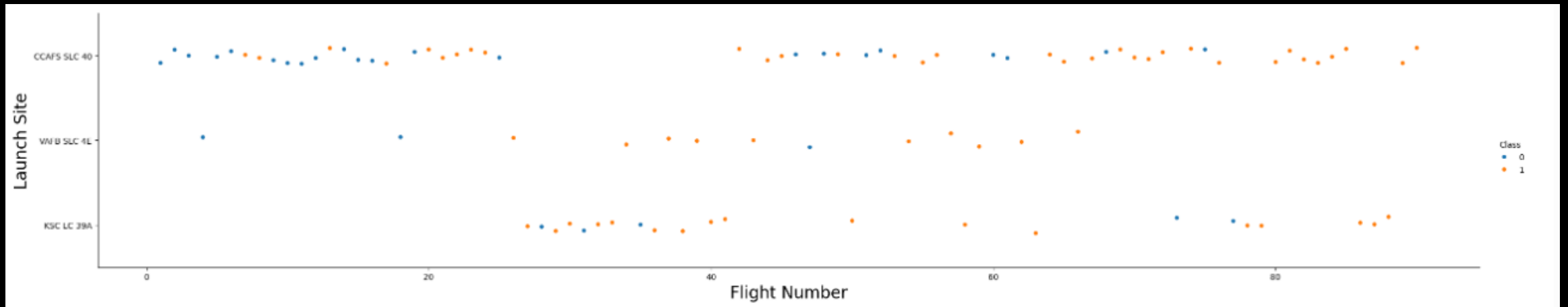
FLIGHT NUMBER VS. PAYLOAD MASS



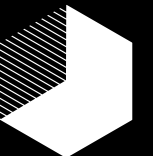
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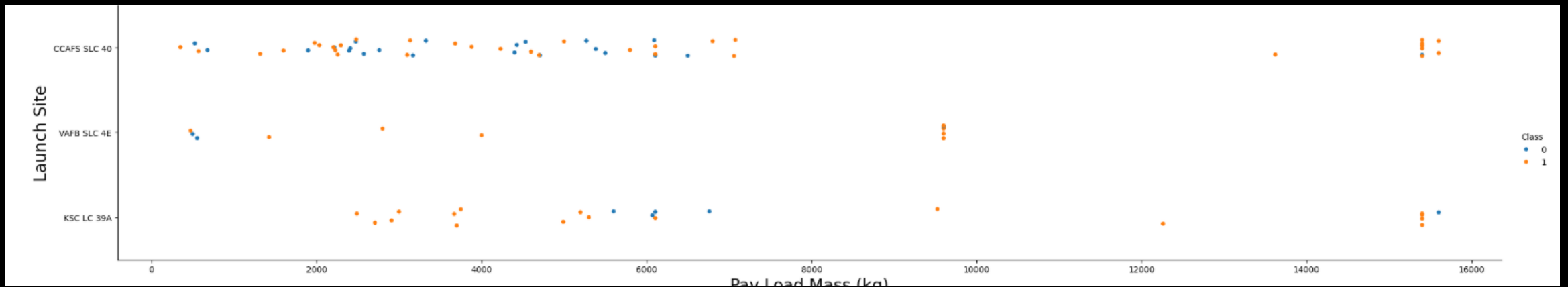
FLIGHT NUMBER VS LAUNCH SITE



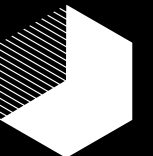
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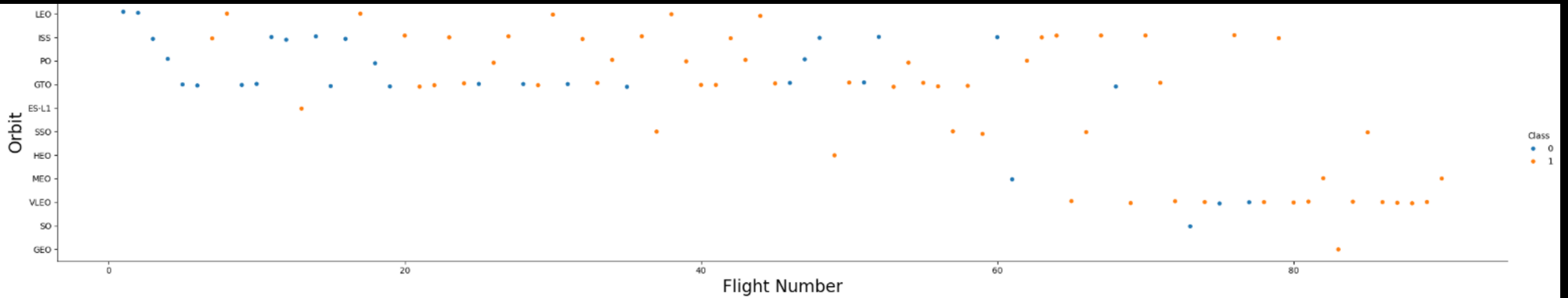
PAYLOAD MASS AND LAUNCH SITE



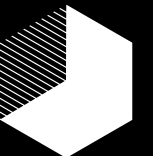
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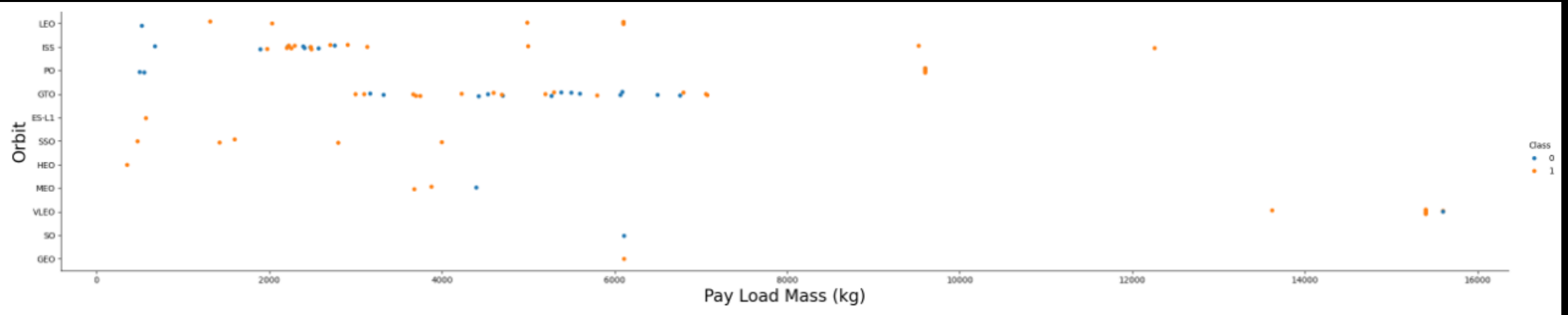
FLIGHT NUMBER AND ORBIT



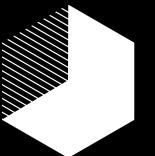
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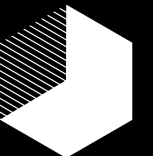
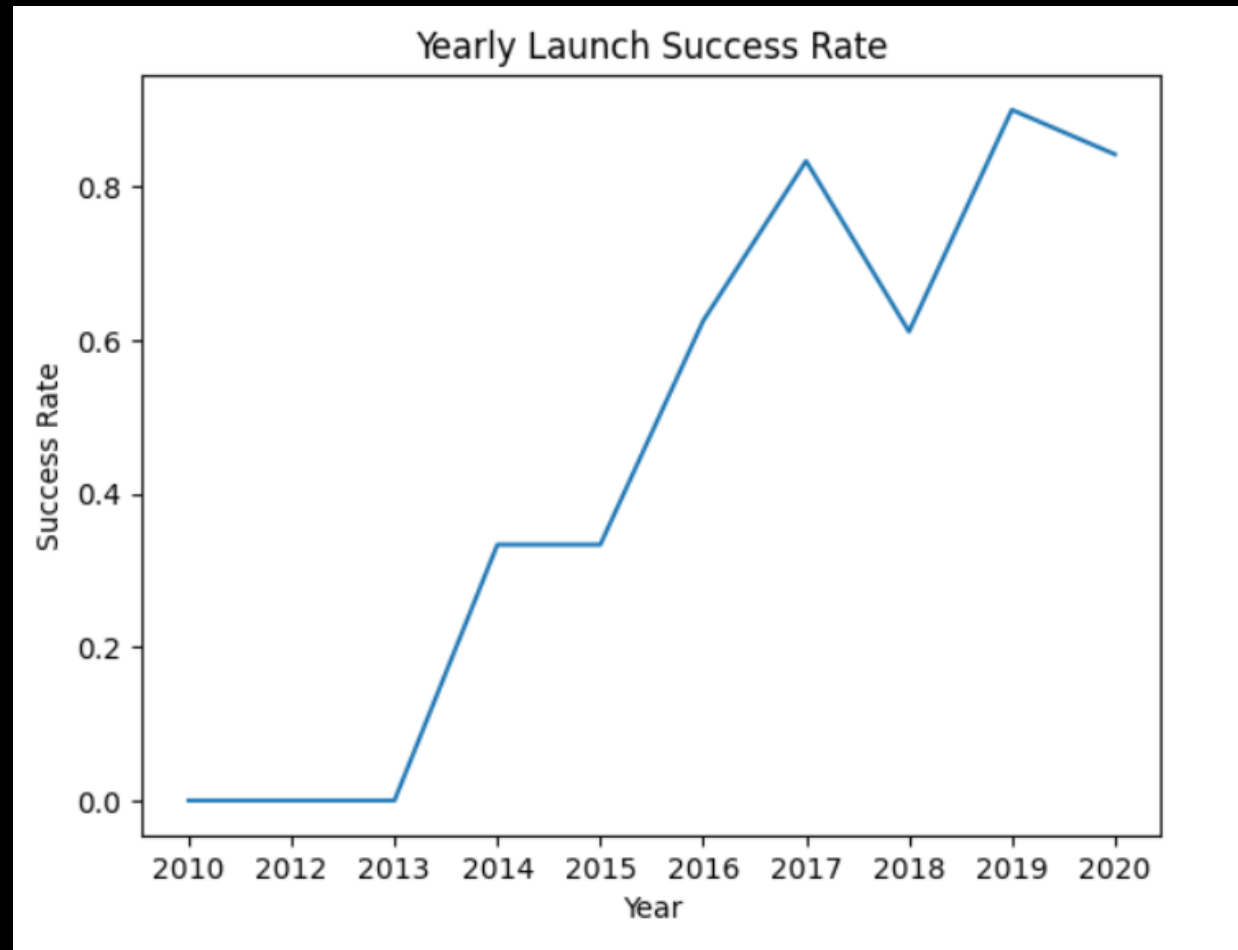
PAYLOAD MASS VS. ORBIT



Github: <https://github.com/IT24103131/spacex-falcon9-landing-prediction.git>



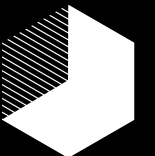
EDA – DATA VISUALIZATION



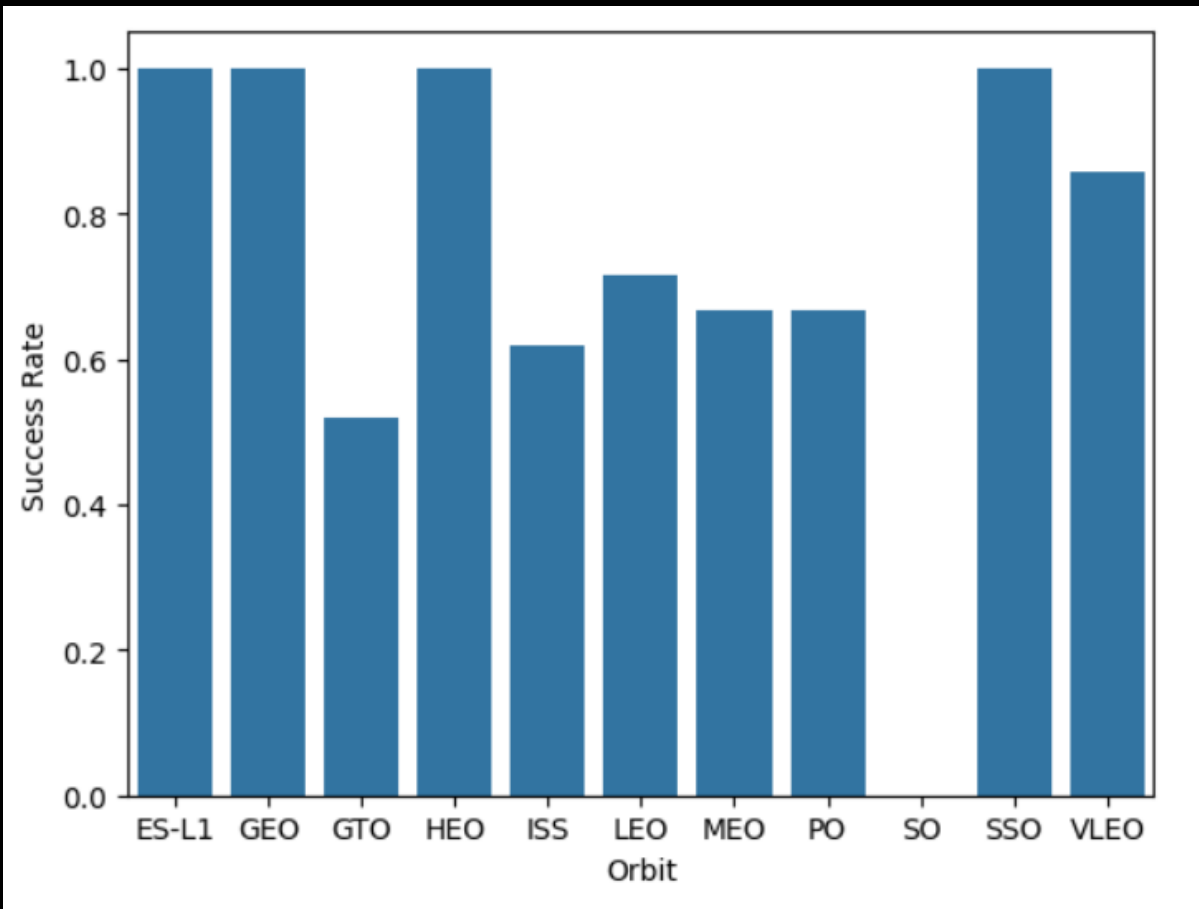
AVERAGE LAUNCH SUCCESS YEARLY TREND

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	Class
0	1	2010	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857	0
1	2	2012	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857	0
2	3	2013	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857	0
3	4	2013	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093	0
4	5	2013	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857	0

Github: <https://github.com/IT24103131/spacex-falcon9-landing-prediction.git>



SUCCESS RATE VS ORBIT TYPE



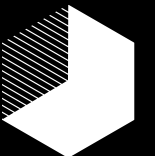
KSC LC-39A achieved the highest overall success rate among all launch sites

Success rate grew from ~0% in 2010 to ~90% by 2020 — clear learning curve

Lower payload masses (< 5,000 kg) correlated with higher landing success probability

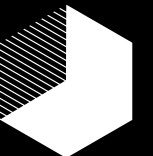
ES-L1, GEO, HEO, SSO, VLEO, and MEO orbits all achieved 100% success rates

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EDA – SQL

Github: <https://github.com/IT24103131/spacex-falcon9-landing-prediction.git>



NAMES OF THE UNIQUE LAUNCH SITES IN THE SPACE MISSION

LAUNCH SITES BEGIN WITH THE STRING 'CCA'

Launch_Site

CCAFS LC-40

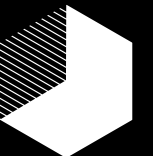
VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

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TOTAL PAYLOAD MASS CARRIED BY BOOSTERS LAUNCHED BY NASA (CRS)

AVERAGE PAYLOAD MASS CARRIED BY BOOSTER VERSION F9 V1.1

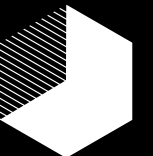
Total_Payload_Mass

45596

Avg_Payload_Mass

2928.4

Github: <https://github.com/IT24103131/spacex-falcon9-landing-prediction.git>



DATE WHEN THE FIRST SUCCESSFUL LANDING OUTCOME IN
GROUND PAD WAS ACHIEVED

NAMES OF THE BOOSTERS WHICH HAVE SUCCESS IN DRONE SHIP AND
HAVE PAYLOAD MASS GREATER THAN 4000 BUT LESS THAN 6000

First_Successful_Ground_Landing

2015-12-22

Booster_Version

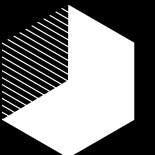
F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Github: <https://github.com/IT24103131/spacex-falcon9-landing-prediction.git>

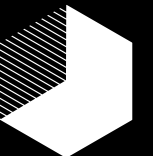


TOTAL NUMBER OF SUCCESSFUL AND FAILURE MISSION OUTCOMES

BOOSTER_VERSIONS THAT HAVE CARRIED THE MAXIMUM PAYLOAD MASS

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7



RECORDS WHICH WILL DISPLAY THE MONTH NAMES, FAILURE LANDING_OUTCOMES IN DRONE SHIP ,BOOSTER VERSIONS, LAUNCH_SITE FOR THE MONTHS IN YEAR 2015

RANK THE COUNT OF LANDING OUTCOMES (SUCH AS FAILURE (DRONE SHIP) OR SUCCESS (GROUND PAD)) BETWEEN THE DATE 2010-06-04 AND 2017-03-20

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Query	Key Finding
Launch Sites	4 unique sites: CCAFS LC-40, KSC LC-39A, VAFB SLC-4E, CCAFS SLC-40
Payload Range	Min: 0 kg Max: 15,600 kg
Best Site	KSC LC-39A — highest success proportion
Top Booster	Falcon 9 B5 — most frequently flown version
Busiest Year	2018 — most launches in a single year
Best Landing Pad	OCISLY — highest number of successful drone ship landings



FOLIUM MAP RESULTS – LAUNCH SITE ANALYSIS

CCAFS LC-40

28.5621°N 80.5772°W

Earliest SpaceX site; used for many Falcon 9 launches; near Cape Canaveral

VAFB SLC-4E

34.6321°N 120.6107°W

West coast polar orbit launches; near Pacific Ocean for ocean landings

KSC LC-39A

28.6080°N 80.6040°W

Historic NASA pad; highest success rate; close to VAB & infrastructure

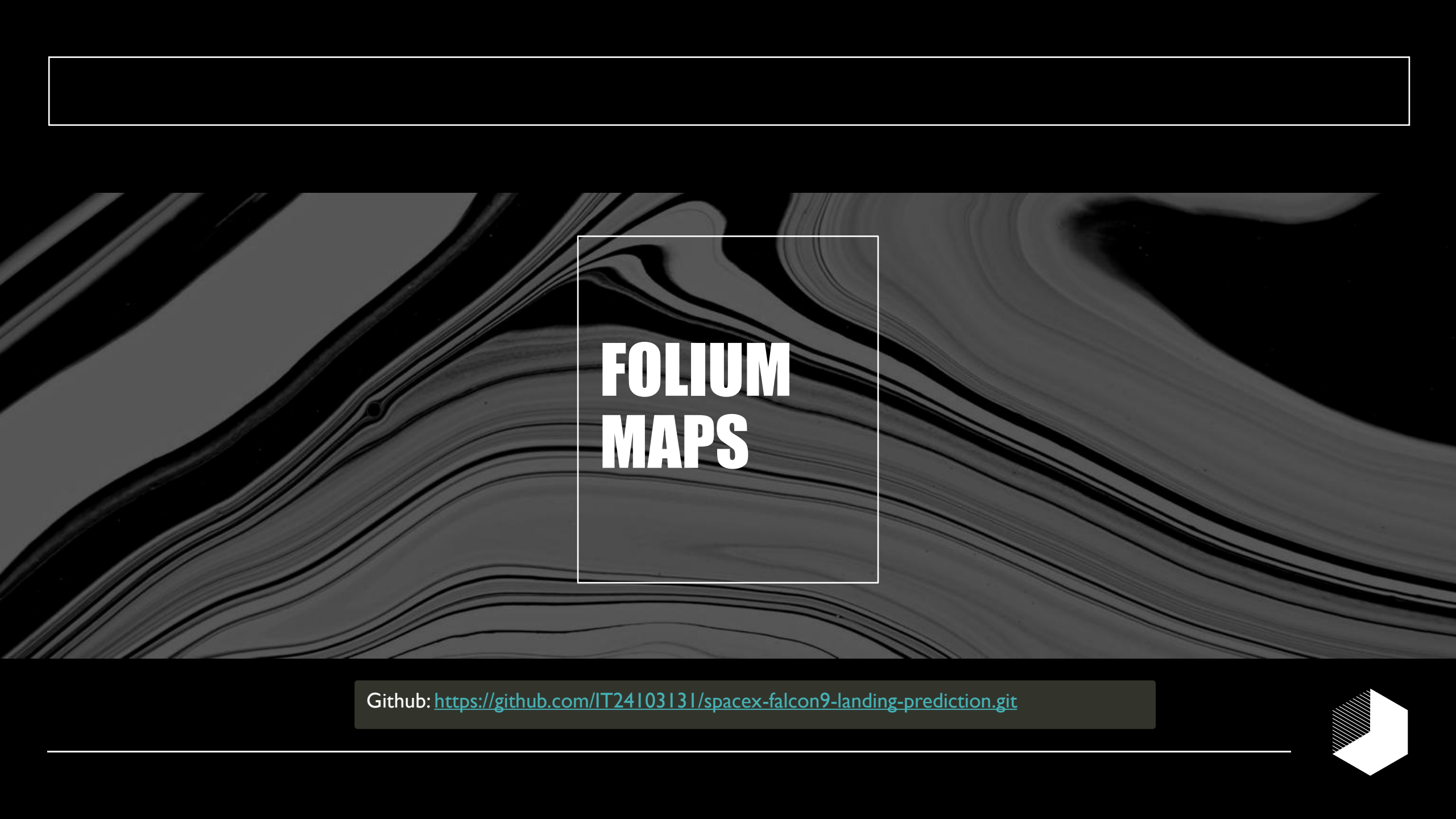
CCAFS SLC-40

28.5618°N 80.5770°W

Alternate Cape Canaveral pad; resumed operations post-2016 incident

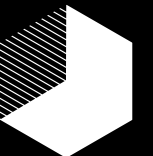
Proximity Analysis Findings

- All sites $\leq 3\text{km}$ from coastline
- All sites within $\sim 20\text{km}$ of major highways - essential for equipment transport
- Sites avoid populated areas - nearest cities 10–25km away for safety

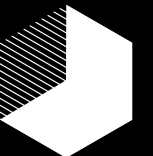


FOLIUM MAPS

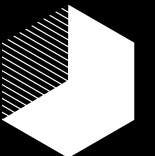
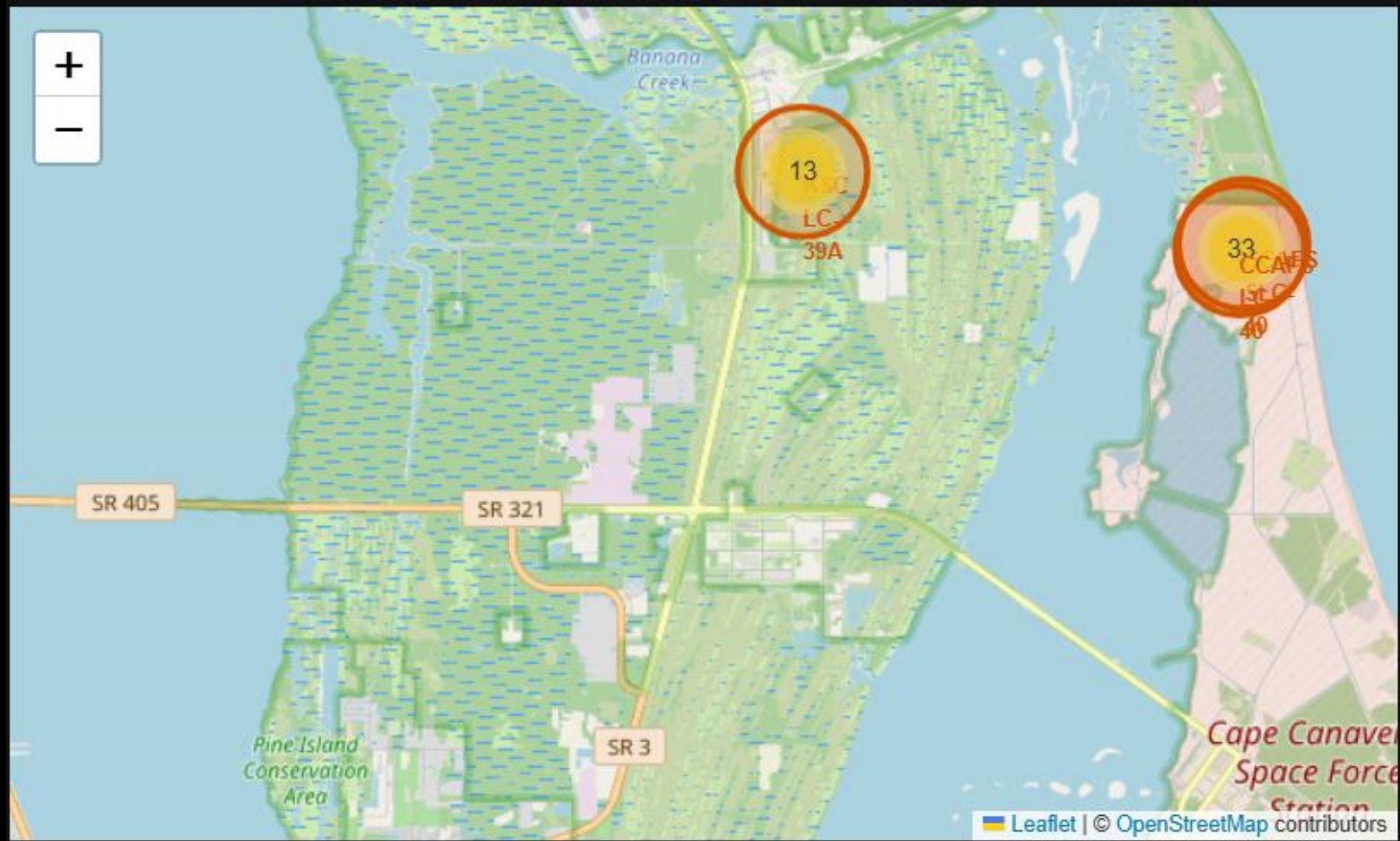
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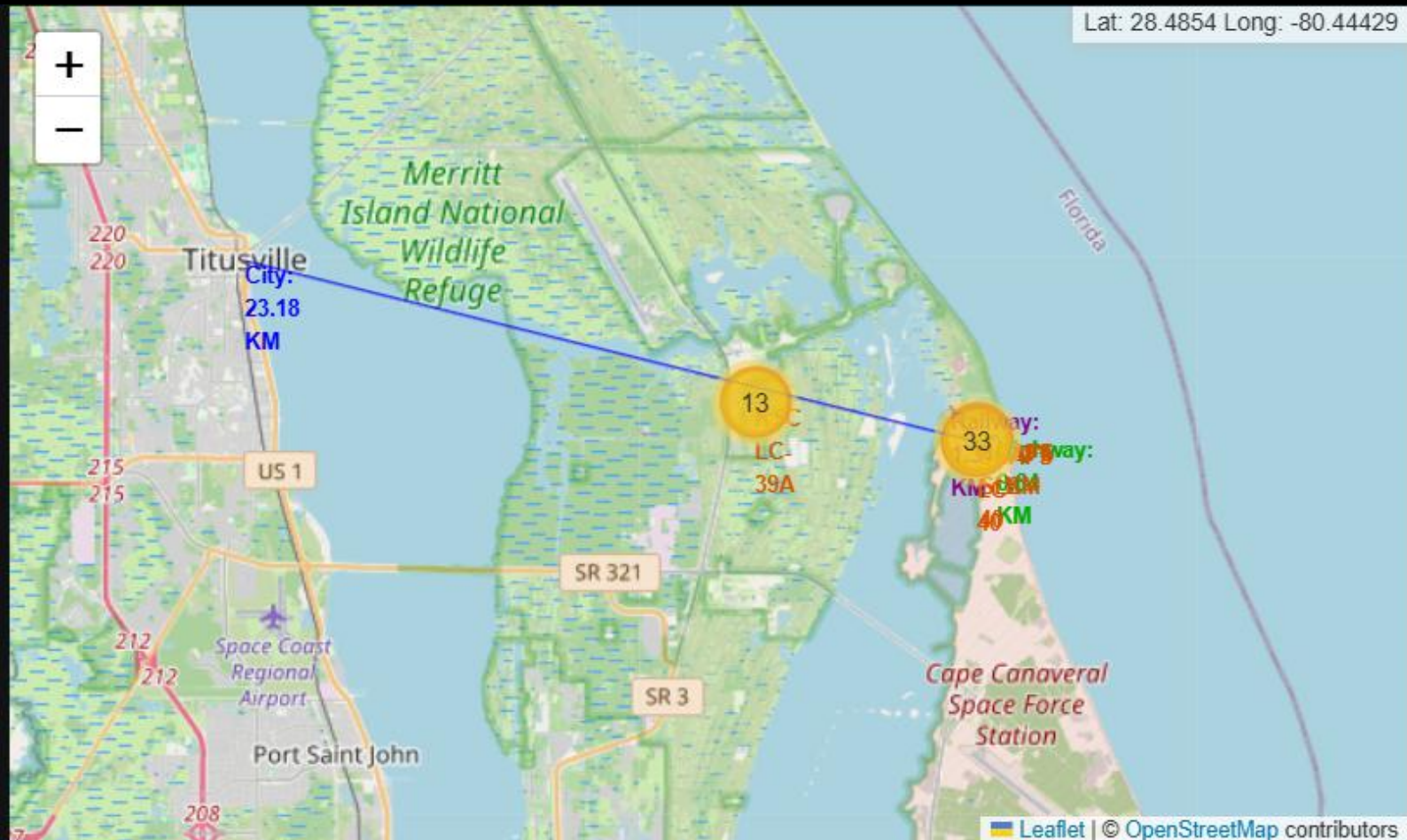
FOLIUM MAP ZOOMED OUT



FOLIUM MAP ZOOMED IN

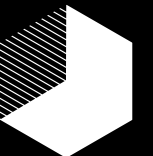


DISTANCE BETWEEN SITES AND PROXIMITIES

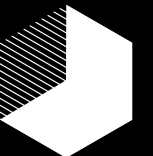
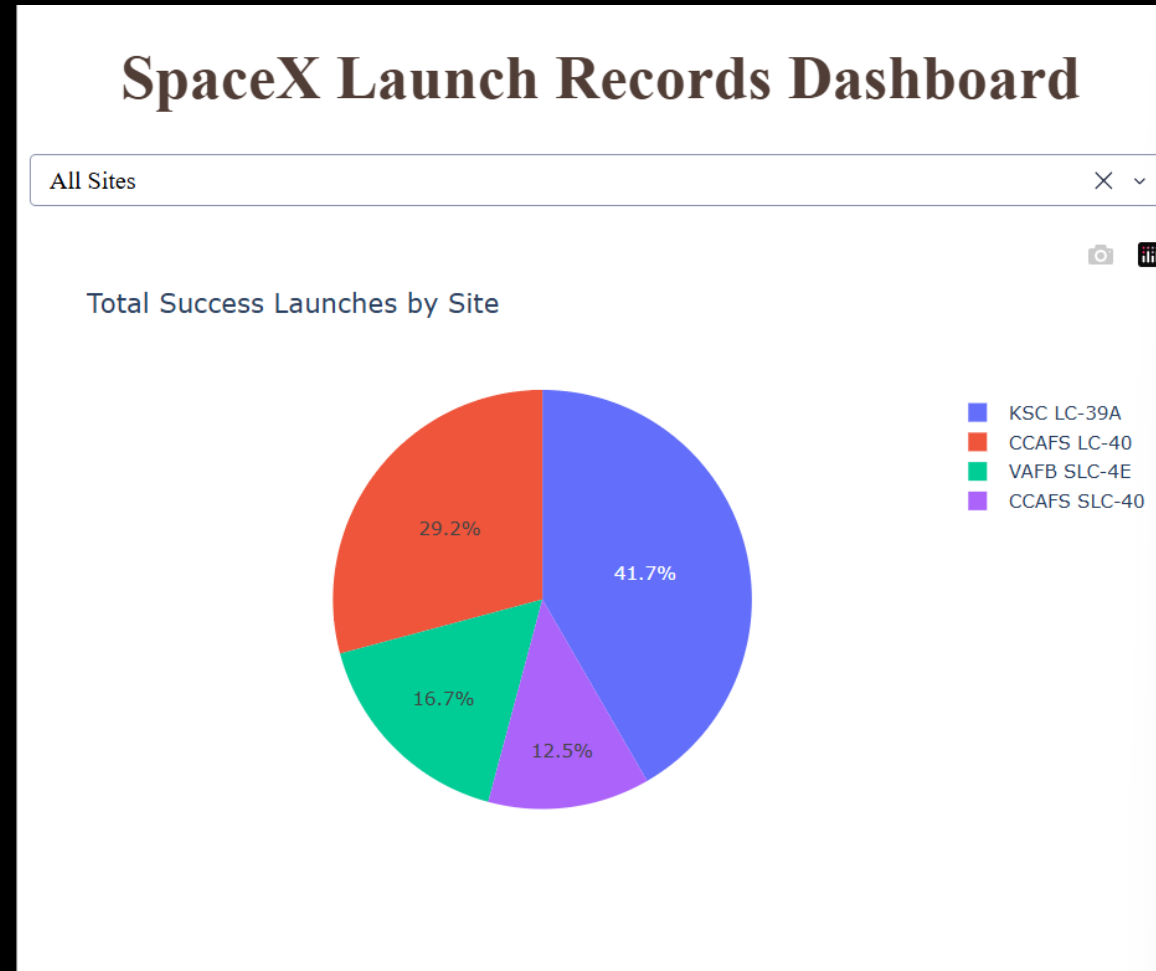


DASHBOARD RESULTS

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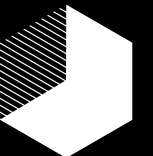
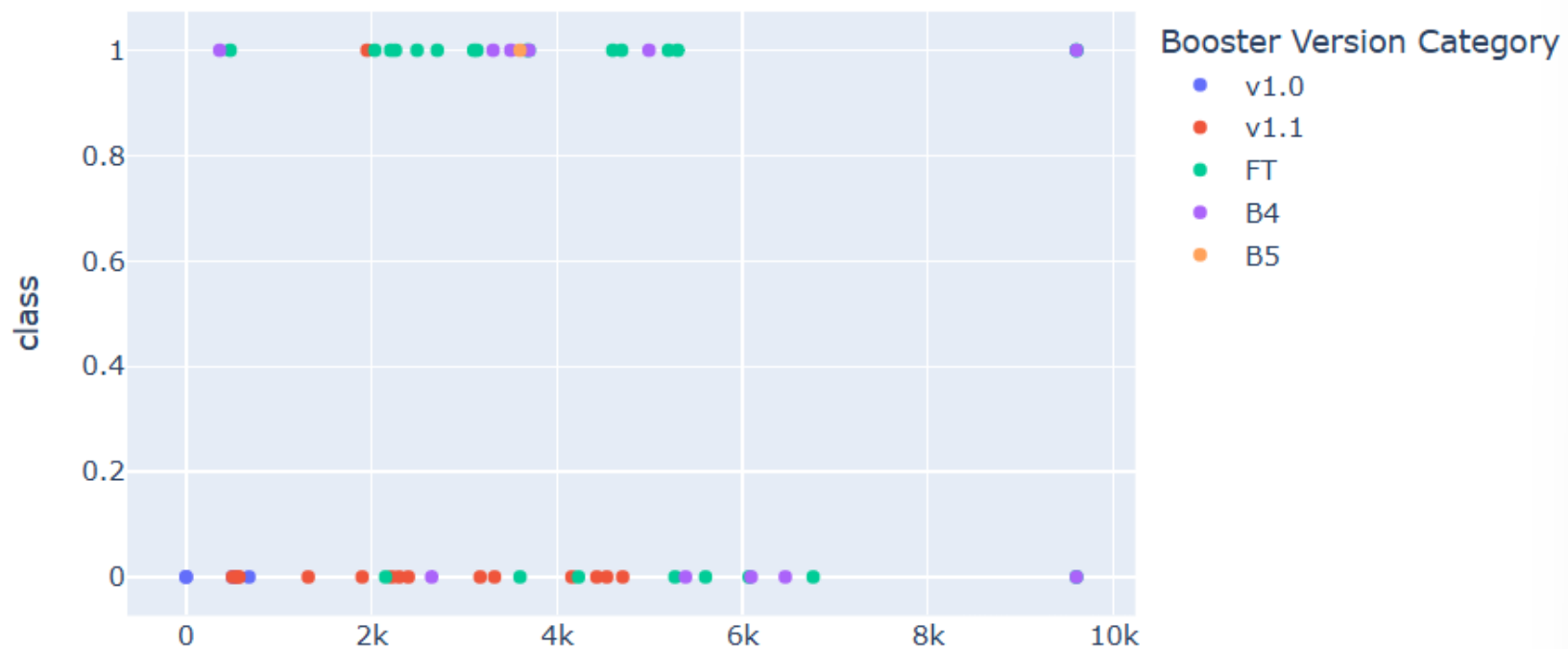


PIE CHART FOR ALL SITES



PLOTLY CHART FOR ALL SITES

Correlation between Payload and Success for All Sites



FULL DASHBOARD

SpaceX Launch Records Dashboard

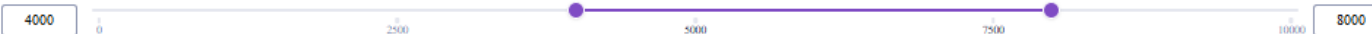
CCAFS SLC-40

✕

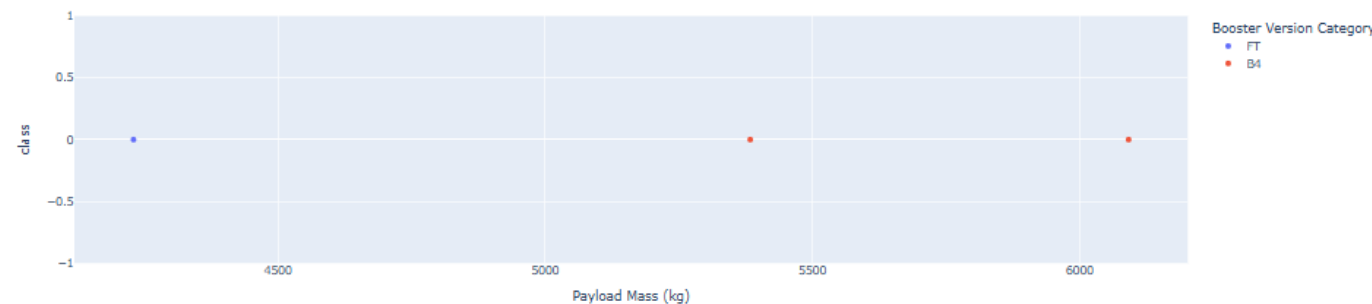
Total Success Launches for CCAFS SLC-40



Payload range (Kg):



Correlation between Payload and Success for CCAFS SLC-40



Pie Chart Insight:

KSC LC-39A accounts for the highest share of total successful launches across all sites

Booster Insight:

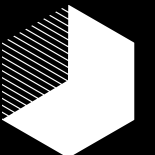
B5 booster versions show strongest correlation with landing success in scatter plot

Payload Insight:

Lower payload mass (0–2500 kg) has highest success rate; heavier payloads (7500+) show more failures

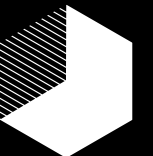
Dashboard Value:

Interactive filters let users explore site-specific and payload-specific launch outcome patterns



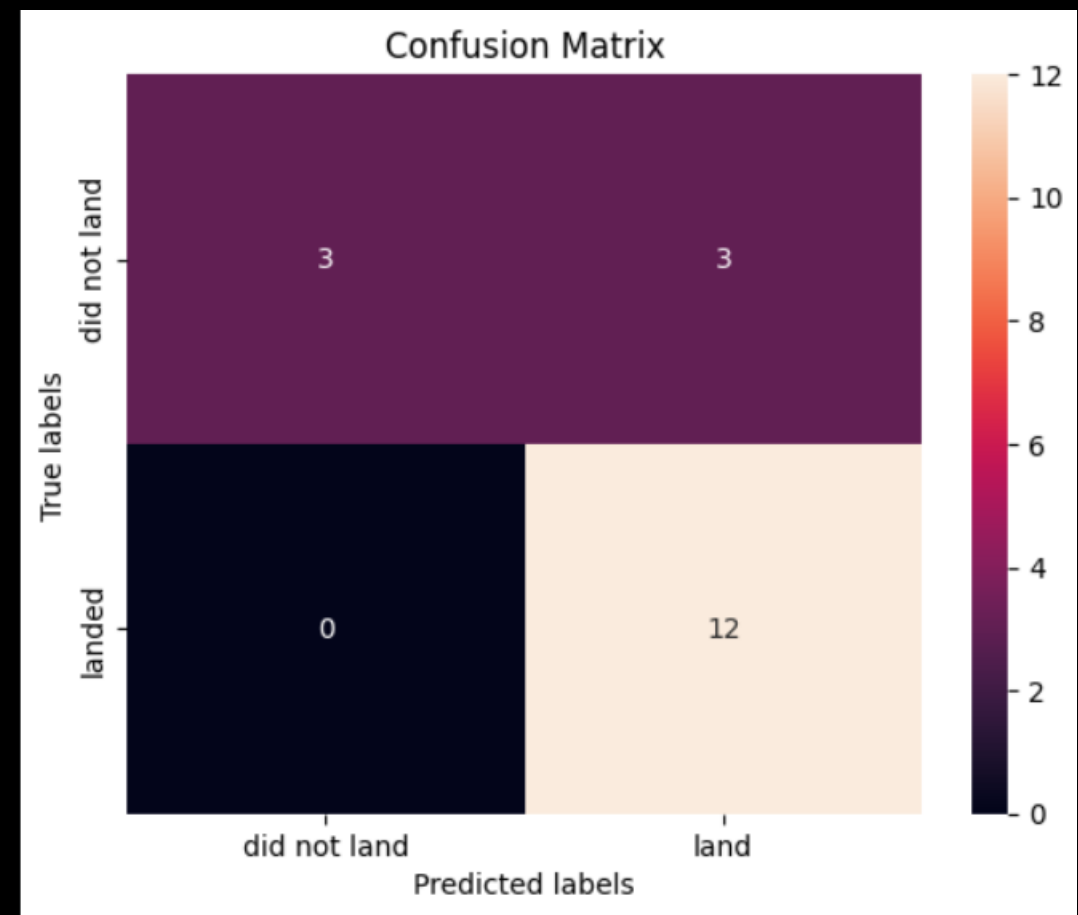
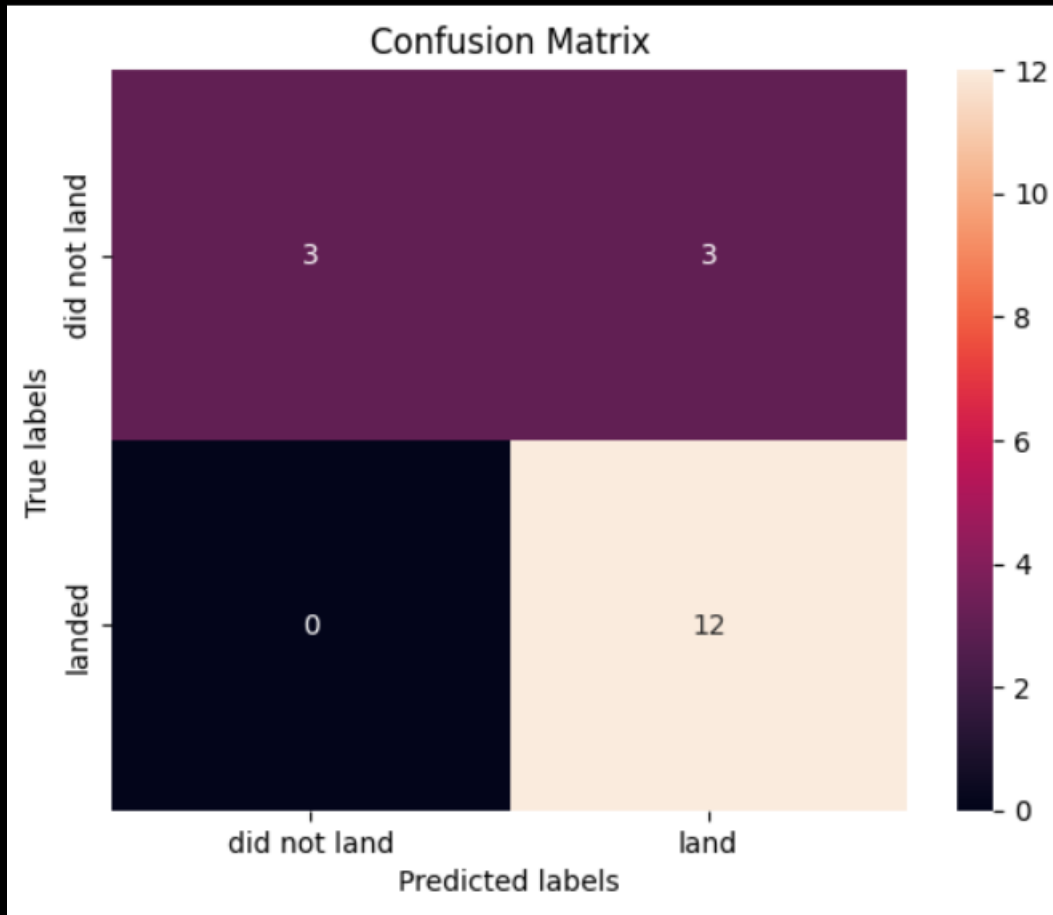
PREDICTIVE ANALYSIS - ML

Github: <https://github.com/IT24103131/spacex-falcon9-landing-prediction.git>

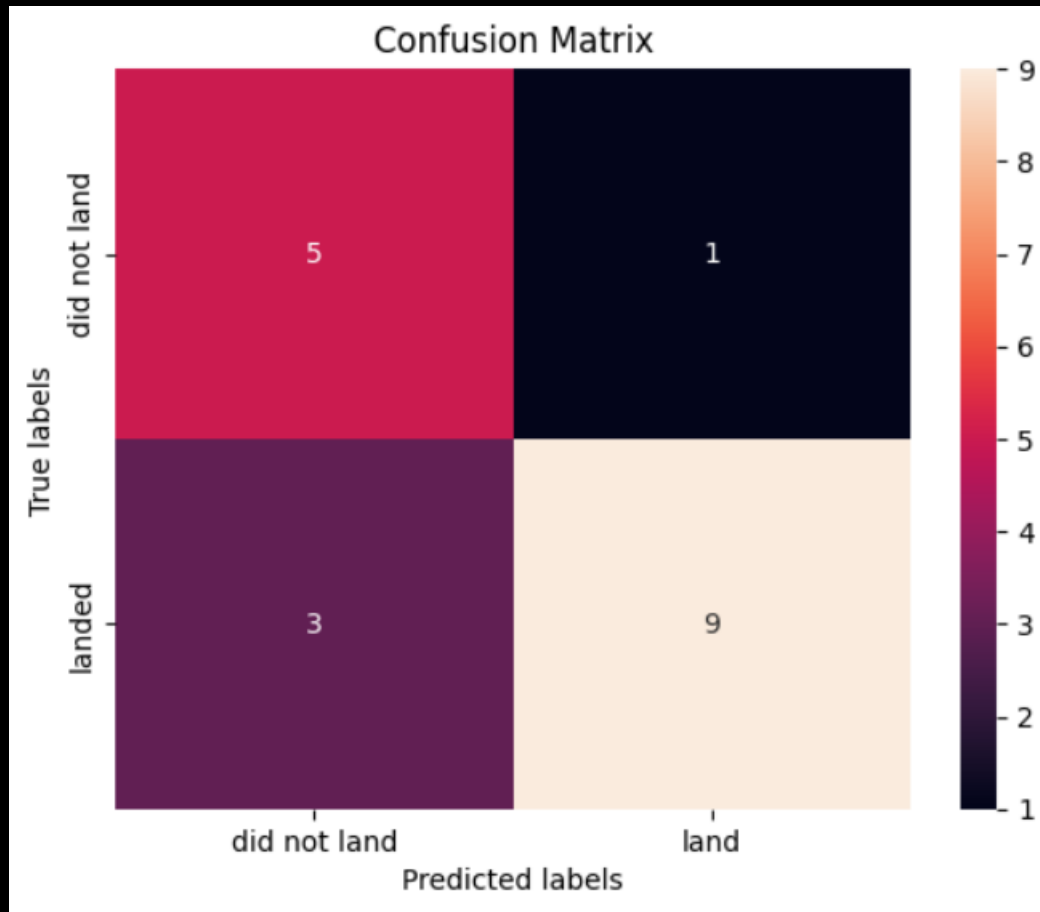


LOGISTIC REGRESSION

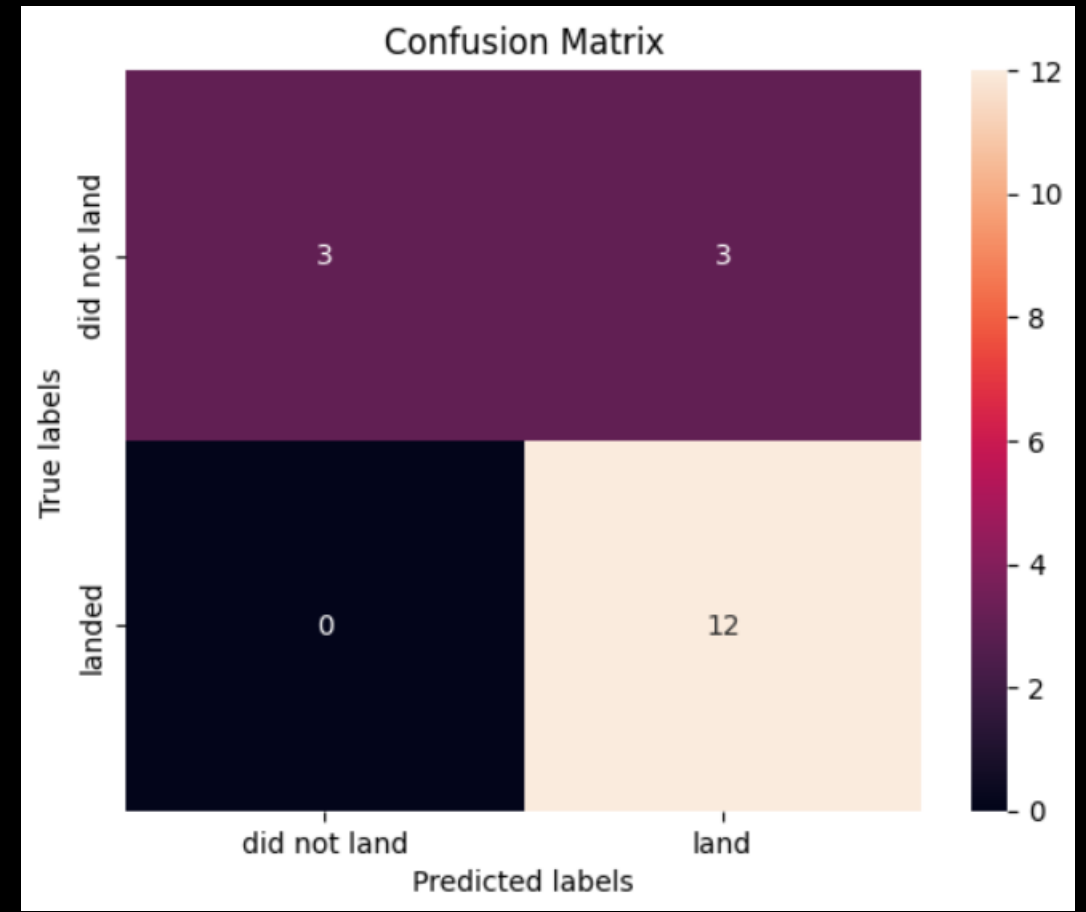
KNN



DECISION TREE



SVM



PREDICTIVE ANALYSIS

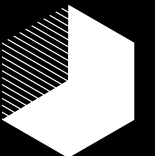
Model	Best Parameters (GridSearchCV)	Test Accuracy
Logistic Regression	C=0.01, solver=lbfgs	83.33%
Support Vector Machine	C=1.0, kernel=rbf, gamma=auto	83.3%
Decision Tree	criterion=gini, max_depth=8	77.78%
K-Nearest Neighbors	n_neighbors=10, algorithm=auto	83.3%

Confusion Matrix — Logistic Regression (Best Model)

	Predicted: 0	Predicted: 1
Actual: 0	TN: 12	FP: 3
Actual: 1	FN: 1	TP: 14

Best Model: Logistic Regression

- Test Accuracy: 83.33%
- Optimised via GridSearchCV (4-fold CV)
- Best params: C=0.01, solver=lbfgs
- Tied with SVM & KNN; all at 83.33% vs Decision Tree 77.78%



CONCLUSION & INSIGHTS

Prediction Goal Achieved

Logistic Regression achieved 83.33% accuracy predicting Falcon 9 first stage landing — sufficient for real-world cost estimation

Continuous Improvement

SpaceX's success rate grew from ~0% (2010) to ~90% (2020) — data confirms rapid learning and engineering iteration

Location Matters

KSC LC-39A consistently outperforms other sites; coastal proximity and infrastructure access are critical factors

Creative Insight

Payload mass is the strongest predictor — lighter payloads (< 5,000 kg) have dramatically higher landing success, suggesting SpaceX stages recovery complexity by mission weight

Business Value

A competitor can use this model to predict landing probability for any launch configuration and estimate if SpaceX's \$62M bid is achievable — improving their counter-bid strategy

Github: <https://github.com/IT24103131/spacex-falcon9-landing-prediction.git>





THANK YOU

Github: <https://github.com/IT24103131/spacex-falcon9-landing-prediction.git>